INCOHST 2026 – Project Proposal

Team Details

1. School Name - Nalanda College - Colombo 10
2. Team Name – Null Pointers
3. Teacher In Charge Name - Dinusha Samanmali
4. Names of team members and Contact Number - (Name to be on Certificate)

   A. Member 1 - Aveesha Ninsara (0784740503)
   B. Member 2-Dinaru Rathnayake (070 175 9585)
   C. Member 3-Lithum Damsithu Ekanayake(071 731 0018)
   D. Member 4-W.A.Sayuru Yukthiya Perera(077 422 7377)

5. Contribution (Each members roles and description) -

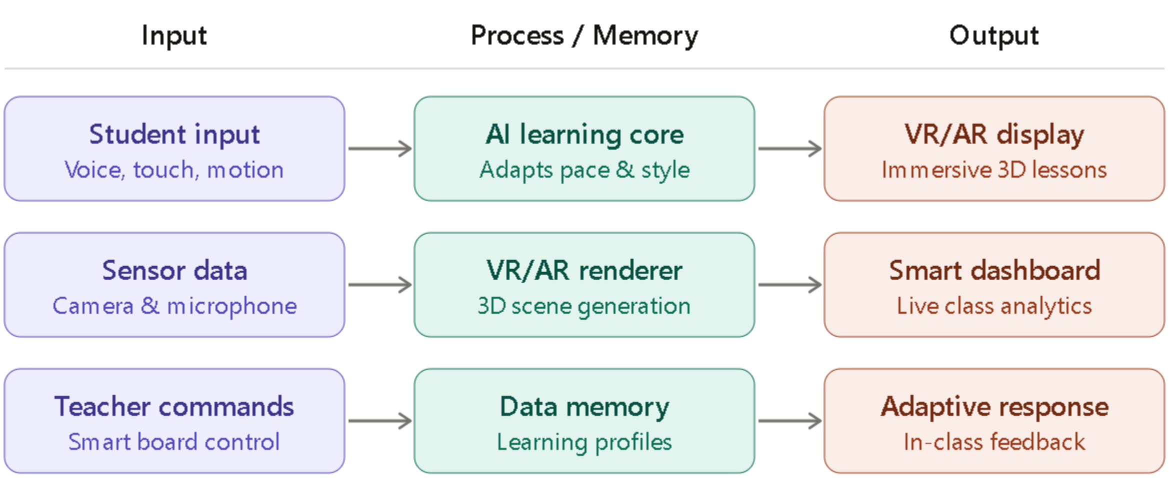
   A. Member 1 - Aveesha Ninsara (Team Leader - Development of the project)
   B. Member 2- Lithum Damsithu Ekanayake(Assistant Team Leader – Software Developer – Development of software side of the project
   C. Member 3- Dinaru Rathnayake (Researcher- researches about the business use-cases, product sustainability and tech stack)
   D. Member 4- W.A.Sayuru Yukthiya Perera (UX/UI Designer & Product Manager – work cooperatively with the software engineer to bring a appealing interface and experience for users by focusing on user flow, wireframes, and aesthetic design. Often acts as the Project Manager to keep the team on schedule and helps with timelines.

Innovation Details

1. Innovation Name- AI Powered Smart Board- A Companion Software with VR and AR
2. Problem Statement-
    a) What we need: Modern classrooms are increasingly failing to sustain student attention and meaningful engagement. In a significant number of schools, particularly those with limited resources, students exhibit visible signs of disengagement during instruction — a problem rooted in two compounding factors. First, conventional teaching methods rely heavily on passive, lecture-based delivery that offers little interactivity or sensory stimulation. Second, core subject content is overwhelmingly presented through static, two-dimensional media — textbooks, diagrams, and whiteboards — that are structurally inadequate for conveying the spatial, dynamic, and relational nature of concepts in science, history, mathematics, and beyond. The cumulative consequence is a dual waste of classroom time for both educators and students, ultimately shifting the burden of true comprehension to unsupported, extended hours of home study.

    b) What we want to achieve: We aim to develop an AI-powered Smart Board companion application that reimagines classroom instruction as an immersive, three-dimensional learning experience. By integrating Virtual Reality and Augmented Reality through an accessible mobile-based VR headset, the platform will deliver subject content in a visually rich and spatially engaging environment — making abstract concepts tangible and exploration-driven. The AI layer will further personalize the learning experience in real time, adapting pacing and complexity to each student's engagement and comprehension. And moreover we are offering users a market place where users can browse, buy, and download VR games, fitness apps, and entertainment experiences directly to users' headsets, By that way we have been available to offer continuous updates to our software and maintain our software making it a free forever application. The goal is to maximize in-class understanding, reduce reliance on supplementary home study, and make high-quality immersive education accessible to schools at all resource levels.

3. Innovation solution Details-
    a) Block diagram of innovation [Input--> Process/Memory-->Output]:



b) Software/Hardware components used in the Innovation:

We're mainly using Expo, built on top of React Native, for our front-end development, along with libraries like React 360, React Native ARKit, and Three.js. For the backend, we're using Node.js. Our hardware stack is fully available in any modern smartphone. The only piece of hardware the user needs to provide is a VR headset. For reference, we use the phone's camera, microphone, and gyroscope as the primary input devices, while the smartphone's screen serves as the main output device in the system.

c) Technical explanation of the Innovation:

The AI Powered Smart Board Companion is a cross-platform mobile application developed using **Expo (React Native)** as the primary front-end framework. The application leverages **React 360** and **Three.js** for real-time 3D scene rendering and immersive VR environment construction, while **React Native ARKit** enables Augmented Reality overlays that anchor virtual objects to real-world surfaces captured through the smartphone camera.

The AI core is powered by a **Node.js backend** that hosts a personalized learning engine. This engine continuously collects interaction data — such as response time, quiz accuracy, and engagement duration — and dynamically adjusts lesson pacing, complexity, and content presentation style to suit each individual student's learning profile. Student profiles and learning history are persisted in a cloud-based database, allowing the system to build longitudinal understanding of each learner's progress.

On the hardware side, the application relies entirely on components available in any modern smartphone: the **camera** serves as the AR input sensor, the **microphone** captures voice commands and verbal responses, and the **gyroscope** tracks head orientation for VR navigation. The smartphone screen acts as the primary display output. When paired with an affordable mobile VR headset, the system delivers a fully immersive stereoscopic 3D learning environment without the need for expensive, dedicated VR hardware.

The Smart Board integration allows teachers to push lesson content, trigger 3D model displays, and monitor a **live analytics dashboard** showing real-time class engagement metrics. An integrated **marketplace module** — built within the same application — enables users to browse, purchase, and download additional VR educational content, fitness experiences, and entertainment applications, generating a sustainable revenue stream that keeps the core educational platform free for all users.

4. Impact of the target group (Social)-

a) Target Group:

- Secondary school students (Grades 6–13) in Sri Lanka, particularly those in under-resourced government and semi-government schools.
- Classroom teachers and subject educators seeking interactive instructional tools.
- School administrators and education policymakers aiming to modernize learning infrastructure cost-effectively.

b) Explanation:

The primary beneficiaries of this innovation are secondary school students, a demographic that is increasingly prone to disengagement in traditional classroom settings due to passive, lecture-driven instruction. Students studying subjects such as Biology, Physics, Chemistry, History, and Mathematics — where spatial and conceptual visualization is critical — stand to gain the most from the immersive 3D learning experiences the platform delivers. By replacing static diagrams and two-dimensional textbook illustrations with interactive, explorable 3D models, students can internalize complex concepts more intuitively and retain information more effectively.

Teachers benefit from real-time engagement analytics displayed on the Smart Board dashboard, empowering them to identify struggling students immediately and adjust their instructional strategy mid-lesson rather than waiting for post-assessment results. This transforms the teacher's role from a passive content deliverer into an informed, data-driven facilitator.

The innovation is also directly relevant to school administrators and policymakers because it delivers a high-quality immersive learning experience at a fraction of the cost of traditional smart classroom infrastructure. Since the system runs on a student's existing smartphone and a low-cost VR headset, there is no requirement for expensive projectors, interactive whiteboards, or dedicated lab computers — making it a scalable and equitable solution for schools across all resource levels in Sri Lanka and beyond.

5. Targeted IOT area and Justify your Selection -
   a) Targeted IoT area: **Smart Education / EdTech IoT**
   b) Justification:

The IoT domain of Smart Education is selected because this innovation fundamentally transforms a conventional physical classroom into a connected, data-aware learning environment — which is the defining characteristic of an IoT-enabled smart space.

The smartphone acts as a multi-sensor IoT node, continuously feeding real-world data — motion, orientation, audio, and visual input — into the cloud-based AI processing layer. This mirrors the standard IoT architecture: sense → transmit → process → respond. The gyroscope and camera feed spatial data into the VR/AR renderer, while microphone input enables voice-driven interaction, collectively turning the student's device into an intelligent, context-aware peripheral.

Furthermore, the AI learning core functions as the intelligent edge-processing layer of this IoT system, making real-time adaptive decisions based on incoming sensor and interaction data — a hallmark of modern IoT intelligence. The Smart Board dashboard aggregates data from all connected student devices simultaneously, functioning as a classroom-wide IoT hub that gives the teacher a unified, live view of the learning environment.

This selection is also justified by the global relevance of Smart Education as an IoT growth area. With educational institutions worldwide investing in connected learning infrastructure post-pandemic, a smartphone-native, VR-integrated platform that requires no proprietary hardware directly addresses the accessibility gap in IoT-powered education — making it both technically sound and socially impactful.

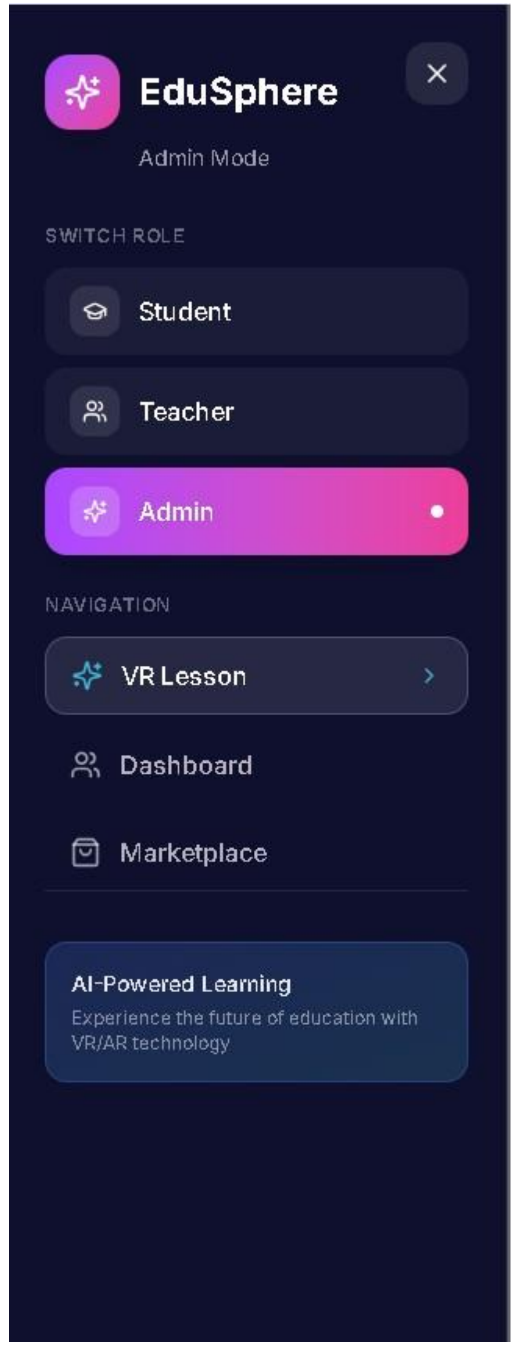
6. Perspective visuals of your innovation (Sketches) –

Dashboard of Our Application.

Marketplace of Our Application.

VR Lesson page of our application.

Currently, this is under development. Once developed, this page will be one of the most important pages of the application, because this will facilitate VR lessons for students with haptic controls.



Page Selector.

Using this, the users can switch between the modes and the pages as the user desires.